Chapter 4 — Descriptive Results

4.1 Introduction

This chapter describes the empirical landscape against which the formal hypothesis tests in Chapter 5 and Chapter 6 are conducted. Its purpose is to characterise the education–fertility relationship in contemporary European populations without imposing parametric assumptions, and to establish the central descriptive finding that motivates the subsequent analysis. I draw on Dataset 1, the panel of education-stratified period education-specific total fertility rates (eTFR) for 21 European countries over 2007–2024 (933 country–year–education observations), and on Dataset 2, the country-level derived dataset used for the cross-national analysis in Chapter 6. The chapter proceeds in four steps: Section 4.2 describes aggregate eTFR trends over the panel period; Section 4.3 characterises cross-country variation in the gradient and its regional patterning; Section 4.4 discusses the distribution of educational attainment and its implications for interpreting low-tier fertility; and Section 4.5 introduces the gradient-shape classification and the core descriptive finding.

4.2 Education-Stratified Fertility Trends, 2007–2024

Across the full panel, period fertility rates remain below replacement in all three educational tiers, consistent with the broader pattern of sub-replacement fertility that has characterised Europe since the 1970s. Table 4.1 presents pooled descriptive statistics for the eTFR by educational tier.

Table 4.1. Descriptive statistics for the education-specific total fertility rate (eTFR) by educational tier, pooled across 21 countries and 2007–2024.

Education tier	Mean eTFR	Min	Max	ISCED codes
Low	1.74	0.36	3.39	0–2
Medium	1.47	0.51	2.61	3–4
High	1.57	0.48	2.42	5–8

Note. eTFR values are education-stratified period total fertility rates computed from age-specific fertility rates in Eurostat data. Education tiers follow ISCED 2011 classifications. N = 933 country–year–education observations.

The most immediate finding in Table 4.1 is that the medium-educated tier records the lowest mean eTFR at 1.47, below both the low-educated tier (1.74) and the high-educated tier (1.57). This ordering — medium is the lowest tier — is the central descriptive finding of this chapter and the primary empirical motivation for the non-monotonicity hypothesis tested in Chapter 5.

The low-education tier shows the widest spread, ranging from 0.36 in the most compressed national context to 3.39 in the highest. This range reflects genuine heterogeneity rather than measurement noise: several Eastern European countries with very low shares of low-educated women aged 25–39 record high eTFR values in that tier, a phenomenon I address in Section 4.4. The high-education tier is markedly more compressed, ranging from 0.48 to 2.42, consistent with the expectation that highly educated women face more homogeneous structural constraints on fertility across diverse national contexts.

Temporal trends over the 2007–2024 window do not show a uniform directional shift in any tier. The panel spans the 2008–2010 financial crisis, the subsequent partial recovery of period fertility in several countries during 2012–2016, and the sharp but short-lived decline associated with the COVID-19 pandemic in 2020–2021. These shocks affect all three education groups but do not appear to alter the cross-tier ordering in any sustained way. The robustness check excluding COVID-affected years (2020–2021) reported in Chapter 5 confirms that the gradient structure is not an artefact of the pandemic period.

4.3 Cross-Country Variation in the SES–Fertility Gradient

Substantial heterogeneity exists in the steepness and direction of the education–fertility gradient across the 21 countries in the sample. The gradient steepness measure used throughout this dissertation is defined as the difference between mean eTFR in the low-education tier and mean eTFR in the high-education tier (eTFR_low − eTFR_high), so that positive values indicate a conventional negative gradient (low educated have more children than high educated) and negative values indicate an inverted gradient. Table 4.3 reports selected country-level values alongside gradient type and share of women in the low-education tier; the full 21-country distribution is available in the appendix.

The gradient steepness values range from +1.322 (Slovakia) to −0.462 (Denmark). Slovakia and Latvia, both with steepness values above 0.6, represent the most extreme conventional negative gradients in the sample; in both countries, low-educated women have approximately twice the fertility of high-educated women. At the other extreme, Denmark is the only country with a clearly inverted gradient on the low–high comparison, with high-educated women recording higher mean eTFR than low-educated women. Belgium approaches inversion but remains close to zero (0.014).

A clear regional pattern is visible. Eastern European countries, including Slovakia, Latvia, Poland, and Czechia, tend to have steeper positive gradients. Nordic countries, particularly Denmark and Norway, show flatter or inverted gradients. Mediterranean countries - Spain, Portugal, and Greece occupy a middle range with moderate positive gradients. This regional variation is consistent with prior descriptive literature on education–fertility differentials in Europe (Skirbekk, 2008; Andersson et al., 2009) and is explored formally in the cross-national analysis of Chapter 6.

4.4 Educational Attainment Distributions and the Interpretation of Low-Tier Fertility

The share of women aged 25–39 in the low-education tier varies substantially across countries, from approximately 6% in several Central European countries to approximately 27% in the highest cases. This variation has important implications for interpreting eTFR values in the low-education tier, because when the low-education group comprises a very small fraction of the female population, its eTFR is likely to be influenced disproportionately by minority populations with compositionally distinct fertility behaviour: recent migrants, Roma communities, and other groups whose fertility may differ systematically from the indigenous low-educated population.

I use a threshold of 12% to distinguish between two types of U-shaped gradient in the classification system described in Section 4.5. Below 12%, the low-education group is sufficiently small that a high eTFR in that tier most plausibly reflects compositional effects rather than a structural feature of the education–fertility relationship for the majority population. Above 12%, the low-education tier is large enough that its high eTFR is more credibly a feature of the general gradient rather than a compositional artefact. This threshold is chosen on substantive grounds and the sensitivity of the main results to alternative values is reported in the robustness section of Chapter 5.

Slovakia exemplifies the composition interpretation: with fewer than 12% of women in the low-education tier and a steepness of 1.322, the high low-tier fertility likely reflects a disproportionate share of Roma women in that group. Romania, by contrast, has both a high share of low-educated women (above 12%) and near-zero gradient steepness (0.004), indicating that the near-absence of a conventional gradient in Romania is a structural feature of that country's fertility landscape rather than an artefact of a small low-education group.

4.5 Gradient Shape Classification and the Predominance of the U-Shape

Drawing on the educational attainment shares and the cross-tier eTFR comparisons, I classify each of the 21 countries into one of four gradient shape categories. The classification uses two sequential decision rules: first, whether high-educated women have higher fertility than medium-educated women (eTFR_high > eTFR_medium); and second, for countries where this condition holds, whether the share of low-educated women falls below the 12% compositional threshold. Table 4.2 presents the full classification.

Table 4.2. Gradient shape classification for 21 European countries based on mean eTFR comparisons and educational attainment share.

Gradient type	N	Countries
Monotonic negative	5	Austria, Portugal, Spain, Sweden, Türkiye
Inverted bottom	2	Belgium, Denmark
U-shape (composition)	8	Croatia, Czechia, Estonia, Finland, Latvia, Poland, Slovakia, Slovenia
U-shape (broad)	6	Greece, Hungary, North Macedonia, Norway, Romania, Serbia

Note. The 12% threshold refers to the share of women aged 25–39 in the low-education (ISCED 0–2) tier. Composition-driven U-shapes have share_low < 12%; broad U-shapes have share_low ≥ 12%. See Section 4.4 for the rationale.

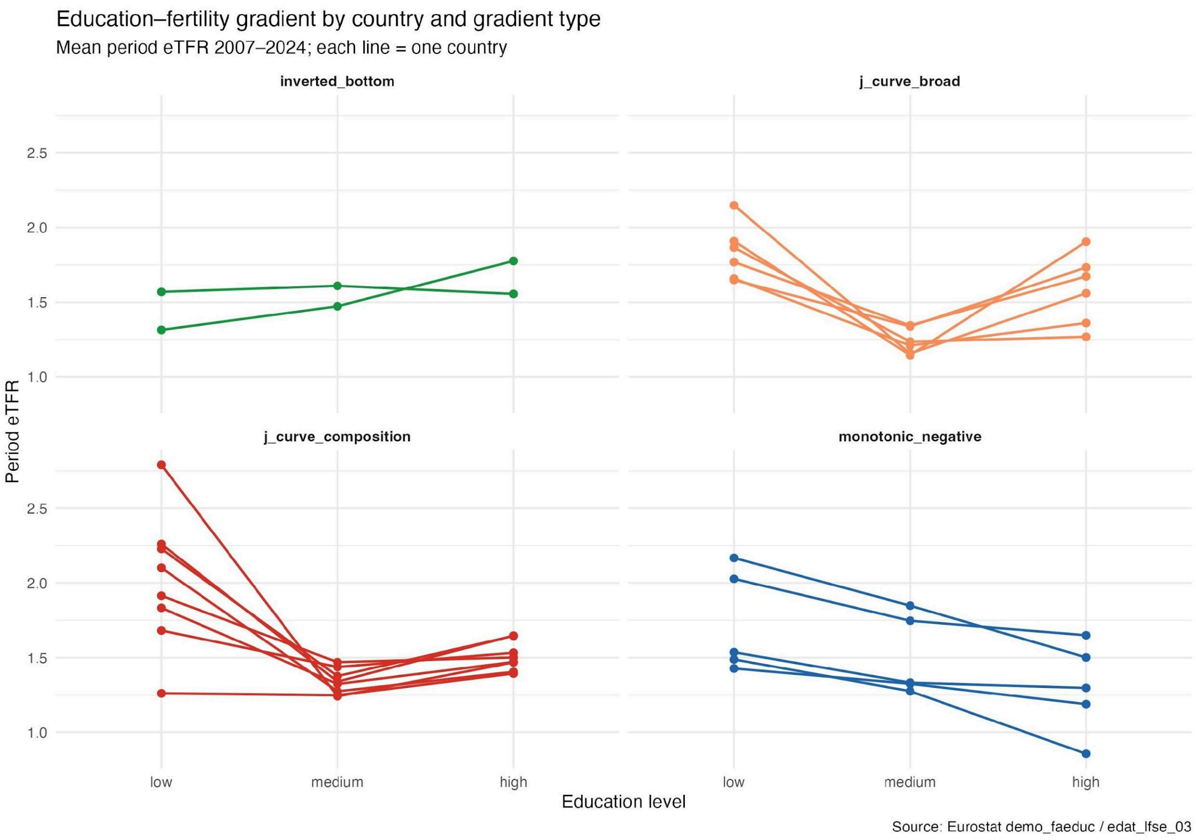

Figure 4.1. Education–fertility gradient by country and gradient type. Mean period eTFR 2007–2024; each line represents one country. Panels show the four gradient-shape categories: inverted bottom (green), U-shape broad (orange), U-shape composition (red), and monotonic negative (blue). Source: Eurostat demo_faeduc / edat_lfse_03.

The most striking feature of Table 4.2 is that 14 of the 21 countries, the combined monotonic-negative and U-shape categories, show higher fertility among low-educated women than among high-educated women, confirming the classical negative gradient at the low–high comparison. However, 12 of these 14 countries (all U-shapes) show medium-educated women recording lower fertility than high-educated women, meaning the classical monotonic pattern (low > medium > high) holds in only five countries in the sample. The dominant pattern is the U-shape, in which medium-educated women sit at the bottom of the fertility distribution.

Figure 4.1 illustrates this visually. The four panels display the gradient profiles for each country grouped by type. The U-shape-composition panel (red) is the most heavily populated and shows a remarkably consistent pattern: steep drops from the low tier to the medium tier, followed by a partial recovery at the high tier. The within-group consistency of this pattern, despite the eight countries spanning Finland in the north to Croatia in the south-east, underscores that the U-shape is a robust empirical regularity rather than a coincidence of a small cluster.

The U-shape-composition group (8 countries) comprises primarily Central and Eastern European countries in which educational upgrading has been rapid over the panel period, compressing the low-education tier and concentrating it among groups with atypically high fertility. Finland is the notable exception in this group, being a Nordic country whose small low-education share nevertheless produces a compositional U-shape. The U-shape-broad group (6 countries), including Greece, Hungary, Norway, and Romania, shows the non-monotonic pattern against a backdrop of substantively large low-education populations, which strengthens the inference that the U-shape is a structural rather than compositional phenomenon in these cases.

Only five countries - Austria, Portugal, Spain, Sweden, and Türkiye — display a strictly monotonic negative gradient across all three tiers, with low > medium > high throughout the panel. Notably, these five countries span diverse regional and cultural contexts, and their common characteristic is a moderate-to-large low-education share combined with a relatively flat relationship between medium and high tiers. Sweden's presence in this group is initially surprising given its high gender-equality scores; however, Sweden's monotonic gradient reflects unusually compressed fertility differences across educational groups overall rather than an exceptionally high low-tier rate.

The two inverted-bottom countries, Belgium and Denmark, show the most divergent pattern: here, the low-education group records lower or similar fertility to the medium group, producing a gradient that rises from low to medium before falling from medium to high (or rising throughout in Denmark's case). The interpretation of Belgium and Denmark requires caution, as both have relatively complex immigration compositions in their low-education tiers.

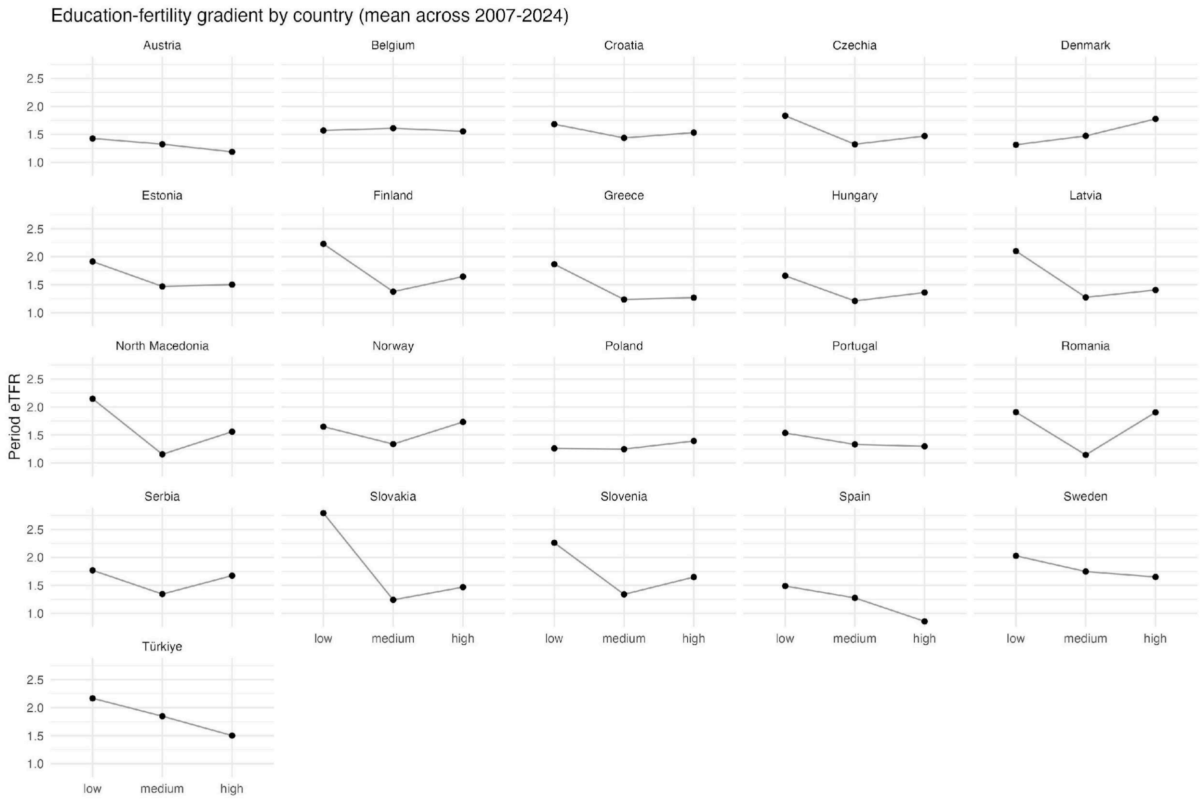

Figure 4.2. Education–fertility gradient by country, mean eTFR across 2007–2024. Each panel shows one country's gradient profile across the three educational tiers (low, medium, high). The dominant U-shape pattern — medium as the lowest tier — is visible in the majority of panels. Source: Eurostat demo_faeduc.

Figure 4.2 disaggregates the gradients to the individual country level, allowing direct visual comparison across all 21 countries. Several features are worth highlighting. Slovakia's profile is the most visually dramatic, with a low-tier eTFR well above 2.5 plunging sharply to the medium tier before recovering modestly. Romania's pronounced V-shape, the steepest drop from low to medium in the sample before a sharp recovery to high, is notable and consistent with the structural broad U-shape classification. Denmark stands out as the sole country whose profile rises monotonically from left to right, confirming its status as the clearest case of gradient inversion in the sample.

Table 4.3. Selected country-level gradient steepness values, gradient shape classification, and share of women aged 25–39 in the low-education tier.

Country	Gradient steepness	Gradient type	share_low (%)	Region
Slovakia	1.322	U-shape (comp.)	< 12	Eastern Europe
Latvia	0.696	U-shape (comp.)	< 12	Eastern Europe
Türkiye	0.666	Monotonic neg.	> 12	Southern Europe
Romania	0.004	U-shape (broad)	> 12	Eastern Europe
Belgium	0.014	Inverted bottom	> 12	Western Europe
Denmark	−0.462	Inverted bottom	< 12	Nordic

Note. Gradient steepness = mean(eTFR_low) − mean(eTFR_high) over the full panel period for each country. Positive values indicate a conventional negative education–fertility gradient. Countries selected to illustrate the range of gradient types; full 21-country table in Appendix A.

4.6 Preliminary Evidence for Non-Monotonicity

The descriptive findings reported in this chapter provide the empirical context for the formal hypothesis tests in Chapter 5. Several features of the data are worth highlighting as preliminary evidence for non-monotonicity before the model-based analysis.

First, the pooled eTFR ordering: medium (1.47) below high (1.57), is not an artifact of a small number of outlier countries or years. Twelve of 21 countries individually show eTFR_medium < eTFR_high over the panel mean, and this pattern is present across the full 2007–2024 window rather than concentrated in specific years. The consistency across country-year cells provides a strong descriptive basis for the non-monotonicity hypothesis.

Second, the country-level distribution of gradient types reveals that no single regional cluster accounts for the U-shape pattern. U-shapes are found in Nordic countries (Norway, Finland), Central European countries (Czechia, Slovakia, Poland), Southern European countries (Greece), and post-socialist Balkan states (Croatia, Serbia, North Macedonia). This distributional breadth argues against an explanation rooted in any single cultural or institutional feature specific to one region.

Third, it is worth noting what the descriptive data do not establish. The U-shape observed at the country-level mean may not hold consistently across all years, all cohorts within the panel, or at the extremes of the education distribution within each tier. It is possible that the U-shape is more pronounced among certain birth cohorts or that it has strengthened or weakened over the 2007–2024 period. It is also possible that the medium-is-lowest pattern, while consistent in the mean, masks quantile-level heterogeneity in which the bottom of the fertility distribution is occupied by high-education women. These questions motivate the quantile regression analysis in Chapter 5, which tests whether the non-monotonic pattern holds across the conditional distribution of fertility, not merely at the conditional mean.

Finally, the descriptive results raise a question about the mechanism through which the U-shape arises. Two interpretations are a priori plausible. Under the first, medium-educated women face the sharpest opportunity costs, they have sufficient human capital to face meaningful wage penalties from childbearing, but insufficient earning power to outsource domestic and childcare labour or to benefit from flexible work arrangements that facilitate fertility among highly educated women. Under the second, the U-shape reflects a compositional feature of the medium-education group, which in many countries includes women in precarious labour market positions for whom fertility postponement translates to completed childlessness. Distinguishing between these interpretations is not possible from the aggregate country-year data alone; it requires the individual-level analysis conducted in

Chapter 6. The descriptive evidence presented here establishes only that the phenomenon is real, consistent, and widespread - sufficient motivation for the formal tests that follow.

4.7 Summary

Four descriptive findings structure the subsequent analysis. First, across the full panel, medium-educated women have the lowest mean eTFR (1.47), below both low-educated (1.74) and high-educated (1.57) women - a U-shaped pattern at odds with the classical monotonic gradient. Second, this U-shape is the dominant pattern in the sample, characterising 12 of 21 countries, while strictly monotonic negative gradients hold in only five. Third, substantial cross-country variation exists in gradient steepness, ranging from +1.322 in Slovakia to −0.462 in Denmark, with a broad regional pattern in which Eastern European countries tend to have steeper conventional gradients and Nordic countries tend to have flatter or inverted gradients. Fourth, the share of women in the low-education tier varies from below 6% to above 27% across countries, and this compositional variation is consequential for interpreting high low-tier fertility in countries with small low-education populations.

These findings motivate the formal model-based tests in Chapter 5. If the U-shape is a genuine feature of the education–fertility relationship rather than a statistical artefact, it should be detectable in fixed-effects models that control for country and year heterogeneity, in quantile regressions that test whether the pattern holds across the conditional distribution, and in semi-parametric models that allow the education–fertility gradient to take a flexible functional form. The results of those tests are reported in the following chapter.